

Fig. 3: Screenshot of configuration and security bits option

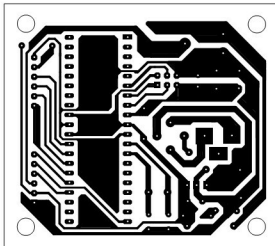


Fig. 4: An actual-size, single-side PCB for the moving-message display on an LCD using AVR microcontroller

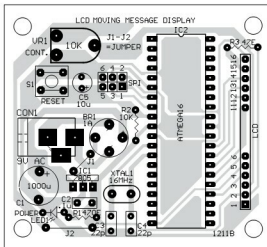


Fig. 5: Component layout for the PCB

Frontline TopView software. This software along with programmer board is available from www.frontline-electronics.com website. After installing the software, select COM port from 'Settings' menu. In 'Device' menu, select the device as ATmega16. Burn

the hex code into the chip by clicking 'Program' option. Note that the microcontroller uses a 16MHz externally generated clock. Program the fuse bits in the software by selecting upper and lower bytes as follows:

Fuse low byte = EF

Fuse high byte = C9

If the fuse bits are not configured properly, the text will scroll slowly or text scrolling may not function properly even if the scrolling rate in the code is changed. The SPI6 connector allows you to program the AVR using SPI adaptor in ISP mode.

Construction and testing

An actual-size, single-side PCB for the moving-message display using AVR is shown in Fig. 4 and its component layout in Fig. 5. Before mounting the AVR onto the PCB, ensure proper power supply to the circuit. Program the AVR as mentioned above, insert it into the PCB and power 'on' the circuit. The text will scroll from right to left in the first line and then in the second line of the LCD, repeatedly. If there is any problem in the display, press reset switch S1

momentarily. If there is no message display at all, vary the contrast-control preset (VR1) until the text is visible on the LCD.

EFY note. All the relevant codes of this article are available on www.efymag.com website.

MOV.M.C

```
/* Program for interfacing a 16x2
LCD module with the ATMEGA16
The program gets the stored text
from program memory of the MCU and then
scroll the text on the LCD.
*/
#define F_CPU 16000000
//16Mhz
#include <avr/io.h> //Header file for AVR
device specific I/O definitions.
#include <avr/pgmspace.h>
//Header file for incorporating
program space string utilities.
#include <avr/interrupt.h>
//Header file for incorporating interrupt
handling facility (not used here).
#include <util/delay.h>
//Header file for incorporating
delay routines.
```

```
#include "lcd2.h"
//Header file for the LCD Module
driver.
#define RATE 250
//Rate of scrolling.
//void LED();
void WaitMs(unsigned int ms);
//Declaration of delay routine used.
char string[] PROGMEM="WELCOME TO ELEC-
TRONICS FOR YOU - NEW DELHI. ";
//
Message stored in the device Flash memory.
int main()
{
    lcd_init(LCD_DISP_ON);
    //LCD Initialisation,
    LCD turned on with cursor off.
    lcd_clear();
    //LCD clear display routine.
```

```
unsigned char len;
//Variable to hold the length of
the message stored.
unsigned char i,k,j=-1;
unsigned char width=16;
//Max width of the LCD display.
for(i=0; {pgm_read_
byte(&string[i])!='\0';i++}); //To calcu-
late the length of the message stored in
the flash memory.
len=i;
while(1) //Unconditional while
loop that set the message scroll continu-
ously on the display.
{
    for(i=15;i!=0;i--)
    {
        j=0;
```